

Zhaolin Gao

CONTACT	zg292@cornell.edu https://zhaolingao.github.io	
EDUCATION	Cornell University ▷ Ph.D. in Computer Science (GPA: 4.09/4.00) ▷ Thesis Committee: Thorsten Joachims, Wen Sun, Claire Cardie ▷ Publications: Google Scholar	Aug. 2023 – Present
	University of Toronto ▷ B.A.Sc. in Computer Engineering (GPA: 3.95/4.00) ▷ Highest academic standing in the final year of the program (1/198)	Sep. 2018 – Jun. 2023
EXPERIENCE	Cornell Tech , New York, NY Cornell University , Ithaca, NY <i>Graduate Student Researcher</i> Collaborating with Prof. Thorsten Joachims and Wen Sun on LLM post-training, reinforcement learning from human feedback, and LLM reasoning. Published at NeurIPS 2024/2025 ($\times 4$), ICLR 2025, and CIKM 2025.	Aug. 2025 – Present Aug. 2023 – Aug. 2025
	Meta Superintelligence , Remote Meta GenAI , Menlo Park, CA <i>Research Scientist Intern</i> Collaborated with Dr. Liang Tan on efficient RL for LLM reasoning through curriculum learning. Published at ICLR 2026.	Aug. 2025 – Dec. 2025 May. 2025 – Aug. 2025
	University of Toronto , Toronto, ON <i>Research Assistant</i> Collaborated with Prof. Scott Sanner on recommendation diversification for variational autoencoder-based recommendation systems. Published at SIGIR 2022.	Nov. 2021 – Jun. 2022
	Layer 6 AI , Toronto, ON <i>Machine Learning Intern</i> Collaborated with Dr. Maksims Volkovs on designing new metric loss objective for collaborative filtering. Published at WWW 2022 and RecSys 2022.	May 2021 – Aug. 2022
	University of Toronto , Toronto, ON <i>Research Assistant</i> Collaborated with Prof. Baochun Li on optimizing training algorithms for Graph Neural Networks under few-shot setting. Published at CVPR 2020 and INFOCOM 2020.	May 2019 – Sep. 2019
AWARDS	LinkedIn PhD Award University Fellowship, Cornell University W. S. Wilson Medal (1/198), University of Toronto 2nd place (2/56), RecSys Challenge 2022 Dean's Honour List, University of Toronto ECE Top Student Award (3/198), University of Toronto The Wallberg Undergraduate Scholarship, University of Toronto First Year Research Fellowship, University of Toronto	2024 2023 2023 2022 2018 ~ 2023 2018 ~ 2020 2018 ~ 2020 2019

PUBLICATIONS **Z. Gao**, Y. Wang, B. Liu, T. Joachims, and W. Sun, *p1: Better Prompt Optimization with Fewer Prompts*, arXiv preprint 2026.

Z. Gao, J. Kim, W. Sun, T. Joachims S. Wang, R. Y. Pang, and L. Tan, *Prompt Curriculum Learning for Efficient LLM Post-Training*, in Proceedings of International Conference on Learning Representations, 2026.

Y. Dai, **Z. Gao**, Y. Sattar, S. Dean, and J. J Sun *Pre-trained Large Language Models Learn Hidden Markov Models In-context*, in Proceedings of the Advances in Neural Information Processing Systems, 2025.

($\alpha - \beta$) K. Brantley, M. Chen, **Z. Gao**, J. D. Lee, W. Sun, W. Zhan, and X. Zhang, *Accelerating RL for LLM Reasoning with Optimal Advantage Regression*, in Proceedings of the Advances in Neural Information Processing Systems, 2025 (also in New York Reinforcement Learning Workshop 2025 **oral**).

K. Wang*, J. P. Zhou*, J. Chang*, **Z. Gao**, N. Kallus, K. Brantley, W. Sun, *Value-Guided Search for Efficient Chain-of-Thought Reasoning*, in Proceedings of the Advances in Neural Information Processing Systems, 2025.

J. P. Zhou, K. Wang, J. Chang, **Z. Gao**, N. Kallus, K. Q Weinberger, K. Brantley and W. Sun, *Q $\ddagger$: Provably Optimal Distributional RL for LLM Post-Training*, in Proceedings of the Advances in Neural Information Processing Systems, 2025 (also in New York Reinforcement Learning Workshop 2025).

Z. Gao, J. Zhou, Y. Dai and T. Joachims, *End-to-end Training for Recommendation with Language-based User Profiles*, in Proceedings of Conference on Information and Knowledge Management, 2025 (also in Workshop on Risks, Opportunities, and Evaluation of Generative Models in Recommender Systems at RecSys'24).

Z. Gao, W. Zhan, J. D. Chang, G. Swamy, K. Brantley, J. D. Lee and W. Sun, *Regressing the Relative Future: Efficient Policy Optimization for Multi-turn RLHF*, in Proceedings of International Conference on Learning Representations, 2025.

Z. Gao, J. D. Chang, W. Zhan, O. Oertell, G. Swamy, K. Brantley, T. Joachims, J. A. Bagnell, J. D. Lee and W. Sun, *REBEL: Reinforcement Learning via Regressing Relative Rewards*, in Proceedings of the Advances in Neural Information Processing Systems, 2024 (also in Models of Human Feedback for AI Alignment Workshop and Aligning Reinforcement Learning Experimentalists and Theorists Workshop (**oral**) at ICML 2024, Workshop on Reinforcement Learning Beyond Rewards at RLC 2024).

Z. Gao, K. Brantley and T. Joachims, *Reviewer2: Optimizing Review Generation Through Prompt Generation*, arXiv preprint 2024.

Y. Lu, **Z. Gao***, Z. Cheng*, J. Sun*, B. Brown, G. Yu, A. Wong, F. Pérez and M. Volkovs, *Session-based Recommendation with Transformer*, in Proceedings of the Recommender Systems Challenge, 2022.

Z. Gao, T. Shen, Z. Mai, M. R. Bouadjenek, I. Waller, A. Anderson, R. Bodkin and S. Sanner, *Mitigating the Filter Bubble while Maintaining Relevance: Targeted Diversification with VAE-based Recommender Systems*, in Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2022.

Z. Gao*, Z. Cheng*, F. Pérez, J. Sun and M. Volkovs, *MCL: Mixed-Centric Loss for Collaborative Filtering*, in the Proceedings of the ACM Web Conference, 2022.

W. Lin, **Z. Gao** and B. Li, *Shoestring: Graph-Based Semi-Supervised Classification With Severely Limited Labeled Data*, in the Proceedings of the 2020 IEEE Conference on Computer Vision and Pattern Recognition, 2020.

W. Lin, **Z. Gao** and B. Li, *Guardian: Evaluating Trust in Online Social Networks with Graph Convolutional Networks*, in the Proceedings of IEEE INFOCOM, 2020.

SERVICES

Reviewer

- ▷ 2026: ICLR, KDD, WWW
- ▷ 2025: NeurIPS, KDD, WWW, ICML EXAIT, ICML MoFA, ICML AI4MATH
- ▷ 2024: TKDE, ICML ARLET, ICML MHFAIA
- ▷ 2022: TOIS

Teaching

- ▷ Cornell Tech, Teaching Assistant, CS 5788 Generative Models
- ▷ Cornell University, Teaching Assistant, CS 4780 Introduction to Machine Learning

Talk

- ▷ *Autoregressive Language Models*, Cornell University, Guest Lecture, CS 5788 Generative Models
- ▷ *Accelerating RL for LLM Reasoning with Optimal Advantage Regression*, New York Reinforcement Learning Workshop 2025
- ▷ *REBEL: Reinforcement Learning via Regressing Relative Rewards*, Cornell University, Guest Lecture, CS 6789 Foundations of Reinforcement Learning
- ▷ *End-to-end Training for Recommendation with Language-based User Profiles*, Workshop on Risks, Opportunities, and Evaluation of Generative Models in Recommender Systems at RecSys'24

Blogpost

- ▷ *Accelerating RL for LLM Reasoning with Optimal Advantage Regression*, Deeper Learning Blog, Kempner Institute, Harvard University [\[Link\]](#)
- ▷ *RLHF 101: A Technical Tutorial on Reinforcement Learning from Human Feedback*, Machine Learning Blog, Carnegie Mellon University [\[Link\]](#)
- ▷ *RLHF 101: A Technical Dive into RLHF*, Huggingface [\[Link\]](#)

Content Creator

- ▷ Bilibili, 29.0k subscribers, 334k likes, 8.9M views [\[Link\]](#)
- ▷ Douyin, 22.4k subscribers, 450k likes, 10M views [\[Link\]](#)
- ▷ YouTube, 1.4k subscribers, 490k views [\[Link\]](#)